

NotesNPU

Private, offline AI lecture & meeting copilot - accelerated by the Snapdragon NPU on HP PCs

Qualcomm AI Hub Whisper on the Hexagon NPU + on-device LLM | 100% offline

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<https://github.com/mdharishsuhaib/NotesNPU>

THE PROBLEM

Taking notes while learning is broken

- Indian students attend 25-30 hours of lectures a week, often in fast, code-mixed Hindi-English, and struggle to take notes and listen at the same time.
- Cloud transcription tools (Otter, Fireflies, cloud Copilots) upload sensitive classroom and meeting audio to third-party servers.
- They need reliable internet and paid subscriptions, which are hard to get in hostels, tier-2/3 colleges, trains and field work.
- Running AI on the CPU drains the battery and heats the laptop, so it can't run through a whole day of classes.

25-30 h

lectures / week per student

0

bytes of audio should leave the laptop

1 day

battery needed for back-to-back classes

NotesNPU: your private AI note-taker

- Record or upload a lecture and get a timestamped transcript, fully on-device, even in airplane mode.
- Qualcomm AI Hub Whisper runs on the Hexagon NPU via ONNX Runtime QNN EP, leaving the CPU free and saving battery.
- A local LLM (Phi-3.5-mini INT4) generates a summary, structured notes, flashcards, a quiz and answers questions.
- Auto-extracts action items and deadlines ('exam next Monday', 'read chapter 6').
- Supports 10+ Indian languages, with Hindi lecture to English notes translation in the same pass.

Transcribe

Upload or record; pause-aware 30 s chunking; timestamps

Live mode

Mic streaming, text every ~8 s, on-device

Translate

Hindi / Tamil / Telugu / Bengali... to English

Study notes

Summary, key concepts, detailed notes, deadlines

Revise

Flashcards and 5-question MCQ quiz

Ask the lecture

Grounded Q&A chat on your transcript

Export

One-click Markdown notes

Benchmark

Built-in NPU vs CPU performance panel

Live demo: lecture in, study notes out

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1. Transcribe 2. Live mode 3. Study notes 4. Ask the lecture 5. NPU benchmark

Lecture / meeting audio

Transcript (timestamped)

Spoken language

English

Translate to English (e.g. Hindi lecture -> English notes)

Load sample lecture

Transcribe on-device

Performance

```

1  {
2    "backend": "ONNX Whisper | CPU",
3    "audio_s": 84.77,
4    "processing_s": 19.82,
5    "real_time_factor": 0.234,
6    "speed_x_realtime": 4.3,
7    "encoder ms avg": 2147.4,

```

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Summary Structured notes Flashcards Quiz

Key Concepts

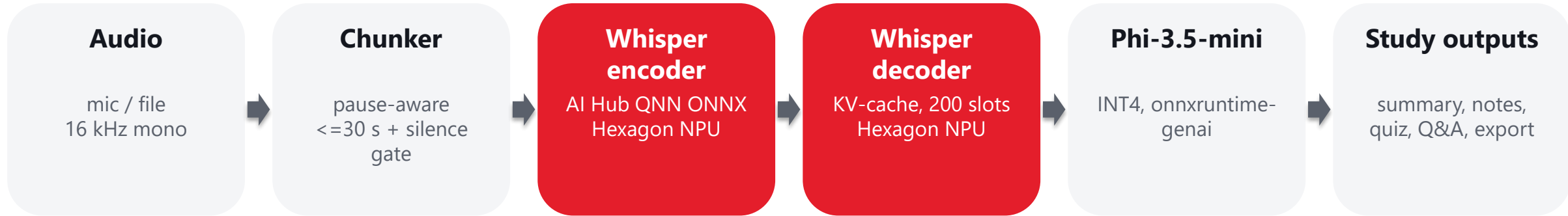
- Photosynthesis:** Process by which plants convert light energy into chemical energy (glucose) using CO₂ and H₂O.
- Chloroplasts:** Organelles within plant cells where photosynthesis occurs, containing chlorophyll.
- Light Dependent Reactions:** Stage involving conversion of light energy into chemical energy (ATP and NADPH).
- Calvin Cycle:** Stages that utilize ATP and NADPH to synthesize glucose from CO₂.
- Factors Affecting Rate:** Light intensity, CO₂ concentration, and temperature influence photosynthesis efficiency.

Detailed Notes

- Location of Photosynthesis:** Mainly in leaf chloroplasts.
 - Occurs in Thylakoid Membranes during Light Dependent Reactions.
 - Splitting Water Molecules: Oxygen released as a byproduct.
 - Produces Energy Carriers: ATP & NADPH.
 - In Stroma during Calvin Cycle.
 - Fixation of CO₂: Formation of Glucose.
- Importance of Photosynthesis
 - Provides Oxygen: Essential for respiration.
 - Foundation of Food Chains: Primary source of organic matter.
- Exam Preparation
 - Draw and Label a Chloroplast: Understand its structure and function.
 - Explain Photosynthesis Significance: Discuss its role in ecosystems.
- Study Material

1. Load or record a lecture > 2. Transcribe on-device (NPU) > 3. Summary / notes / flashcards / quiz > 4. Ask & export

Architecture: everything runs on the PC



- ONNX Runtime QNN Execution Provider (plugin EP, HTP backend, burst perf mode) runs AI Hub precompiled context binaries, so there is no on-device compile.
- Encoder emits cross-attention KV caches directly; the fixed-shape decoder loop feeds self-attention caches back, so every step stays on the NPU.
- Robust by design: NPU to CPU to extractive fallback chain, repetition guard, silence gating, head+tail context fitting for long lectures.

Why the Snapdragon NPU

Task	Model	Source	Runtime / hardware
Speech-to-text (NPU)	Whisper-Base, precompiled QNN ONNX FP16	Qualcomm AI Hub	ONNX Runtime + QNN EP (HTP) on the Hexagon NPU
Speech-to-text (fallback)	Whisper-Base ONNX (INT8 decoder)	Hugging Face onnx-community	ONNX Runtime CPU EP
Study assistant	Phi-3.5-mini-instruct INT4 AWQ	Microsoft (Hugging Face)	onnxruntime-genai

- Whisper encoder = dense transformer over 1500 frames, a good fit for the Hexagon HTP's FP16 tensor engines (measured: 45.5 ms per 30 s of audio on X Elite).
- AI Hub precompiled context binaries: no on-device compile, instant start-up.
- Fixed-shape KV-cache decoder (200 slots) keeps every step on the NPU (3.8 ms/token).
- NPU does the speech, so the Oryon CPU stays free for the LLM and UI and live mode never stutters.
- Low-power NPU inference means a full day of lectures on battery, quiet and cool.

Performance: measured on a real Snapdragon NPU

Device / backend	Speed	Encoder per 30 s	Decoder per token
Snapdragon X Elite, Oryon CPU (measured on Qualcomm AI Hub)	~14x real time (~6.1 s)	1286 ms	9.1 ms
Snapdragon X Elite, Hexagon NPU (measured on Qualcomm AI Hub)	~80x real time (~1.1 s)	45.5 ms	3.8 ms
NPU speed-up	~5.7x end to end	28.3x	2.4x

- Measured on a real Snapdragon X Elite (Qualcomm AI Hub hosted device): encoder 45.5 ms per 30 s of audio, decoder 3.8 ms per token.
- 100% of ops on the NPU: 556 encoder ops + 975 decoder ops on the Hexagon HTP, none on the CPU; peak memory 34 MB / 60 MB.
- Accuracy: NPU (FP16) encoder output matches the FP32 CPU reference with cosine similarity 0.999, and gives an identical, correct transcript of the sample lecture.
- The 85 s sample lecture takes ~1.1 s on the NPU vs ~6.1 s on the same Snapdragon X Elite's Oryon CPU.

Who it helps

Students	Focus on understanding, not scribbling; revise with auto flashcards and quizzes.
Accessibility	Live captions for deaf and hard-of-hearing learners; notes for students with dyslexia or ADHD.
Regional language learners	Hindi / regional lectures turned into English notes, bridging the medium-of-instruction gap.
Professionals	Confidential meetings summarised with action items, and nothing is uploaded.
Teachers	Publish lecture notes and question banks from their own recordings.

Deployment & accessibility

- One-click setup.bat / run.bat: detects Snapdragon, installs native ARM64 deps, downloads AI Hub models.
- Runs as a local web app on 127.0.0.1, with offline mode enforced (HF_HUB_OFFLINE, no analytics).
- Automatic fallback: NPU to CPU to extractive engine, so it works on every Windows PC.
- Light install (-NoLLM, ~300 MB) for low-storage laptops.
- Open source (MIT) on GitHub, with CLI tools for transcription and benchmarking.

```
> git clone <repo>  
> cd notesnpu  
> setup.bat  
> run.bat
```

Opens <http://127.0.0.1:7860>
Works offline after setup

Roadmap

- LLM on the NPU: AI Hub Llama-3.2-3B / Phi-3.5 QNN binaries via Genie
- Speaker diarisation for meetings (who said what)
- Slide / whiteboard capture with on-device OCR, linked to the timeline
- Semantic search across a semester of lectures (on-device embeddings)
- MSIX installer on the Microsoft Store; HP AI Companion integration

Thank you

NotesNPU: learn more, write less, and keep it on your PC.

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